# A Step in the Right Direction: Bridging Forecasting and Classification for Temporal Foundation Models

**Rafael Montano**
The University of Sydney
Business School

**Pablo Montero-Manso**
The University of Sydney
Business School

**Andrey Vasnev**
The University of Sydney
Business School

## Abstract

As temporal foundation models are developed to support a broader range of downstream tasks, a central question is whether forecasting performance translates into task specific predictive utility. Directional prediction—whether a future observation increases relative to the latest observed value—is relevant to decision making in many financial and operational settings. We study it as a shared task connecting time series forecasting and classification. We introduce a framework that reformulates forecasting datasets into horizon specific classification tasks, enabling direct comparison between directions inferred from forecasting foundation models and those predicted by classification foundation models. Experiments across the six M4 Competition frequencies reveal contrasting horizon dependent behaviour. On the Daily subset, directional accuracy derived from the forecasting foundation model Chronos-2 generally decreases with horizon, while that of the classification foundation model Mantis generally increases. A post hoc adjustment analysis further explores whether classifier derived directional signals can complement point forecasts. Together, these findings establish directional prediction as a common ground for evaluating forecasting and classification approaches, motivating broader assessment of temporal foundation models beyond conventional forecast scores.

## 1 Introduction

Directional prediction connects two approaches to anticipating future change: deriving a direction from a numerical forecast and learning the directional target directly from historical observations. Both can inform financial and operational decisions, for which directional accuracy has long been recognised as relevant (Leitch and Tanner, 1991; Granger and Pesaran, 2000). However, forecasting and time series classification methods typically use different benchmark tasks and evaluation criteria, complicating direct comparison.

Recent work investigates whether forecasting foundation model representations transfer to time series classification (Auer et al., 2025). We examine a complementary connection: applying classification approaches to predict future directional changes on forecasting datasets. This shared task offers a concrete test of temporal foundation models' utility beyond conventional forecast scores.

We ask: *how do forecast derived and directly learned directional predictions compare across prediction horizons?* Our Directional Prediction Framework assigns horizon specific labels indicating whether a future observation increases relative to the forecast origin. We compare forecasting and classification methods across the six M4 Competition frequencies (Makridakis et al., 2018), including Chronos-2 forecasts (Ansari et al., 2025) and classifiers using Mantis representations (Feofanov et al., 2025). Direction is clearly defined, yet predicting it in real world applications remains challenging. A post hoc adjustment analysis explores whether Mantis derived signals can complement point forecasts.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Foundation Models for Temporal Systems.

Our contributions are threefold: **(i)** a shared framework for comparing forecasting and classification approaches on directional prediction; **(ii)** descriptive evidence of differences across methods, frequencies, and horizons, including contrasting Daily horizon profiles; and **(iii)** an exploratory analysis of the forecasting value of classifier derived directional signals.

## 2 Related Work

**Directional accuracy** measures whether a prediction correctly identifies the direction of change of a target variable. It has long been recognised as an important dimension of forecast evaluation because many decisions depend more on anticipating future movements than on minimising point forecast error (Leitch and Tanner, 1991; Granger and Pesaran, 2000). Despite its practical importance, directional prediction has traditionally been treated as a secondary outcome of point forecasting rather than as a task in its own right (De Gooijer and Hyndman, 2006; Hyndman and Koehler, 2006). Recent studies underscore the need for precise directional evaluation protocols, as ambiguous formulations can distort directional accuracy estimates (McCarthy and Snudden, 2025).

Recent advances in **time series classification** (TSC) have produced effective methods for extracting or learning discriminative temporal representations, including ROCKET and InceptionTime, together with standardised evaluation frameworks based on benchmark repositories such as the UCR Archive (Bagnall et al., 2017; Dau et al., 2019; Dempster et al., 2020; Ismail Fawaz et al., 2020). At the same time, **time series foundation models** have created new opportunities for forecasting and classification. Chronos-2 uses group attention for zero shot univariate, multivariate, and covariate informed forecasting (Ansari et al., 2025). Mantis is a lightweight transformer based foundation model for time series classification trained with self supervised contrastive learning (Feofanov et al., 2025).

These developments motivate a unified view of directional prediction. Although point forecasting methods and time series classifiers, including foundation model based approaches, all capture temporal information, they are typically evaluated using different criteria and therefore remain difficult to compare directly. Our framework enables their comparison using directional accuracy as a common evaluation criterion across forecast horizons.

## 3 Method

We introduce a Directional Prediction Framework that reformulates each forecasting task as a binary classification problem: predicting whether a future observation increases relative to the forecast origin. The framework evaluates point forecasting methods and time series classifiers using directional accuracy. When the classifier and forecast imply different labels, the classifier label determines whether the upward or downward adjustment factor is applied to the point forecast.

**Forecasting notation.** Let $\mathbf{y} = (y_1, y_2, \ldots, y_T)$ be the observed time series, with forecast origin $T$ and final observed value $y_T$. At horizon $h$, $y_{T+h}$ is the future observation and $\hat{y}_{T+h|T}$ is its point forecast. Let $H$ denote the maximum forecast horizon, with $h \in \{1, \ldots, H\}$.

**Binary sign convention.** Following McCarthy and Snudden (2025), define $\mathrm{sign}(y) = 1$ if $y > 0$, and $\mathrm{sign}(y) = -1$ if $y \leq 0$.

**Directional accuracy.** At forecast origin $T$ and horizon $h$, directional accuracy is defined as

$$d_{T,h} = \mathbb{I}\left[\mathrm{sign}(y_{T+h} - y_T) = \mathrm{sign}(\hat{y}_{T+h|T} - y_T)\right]. \tag{1}$$

Both changes are evaluated relative to the forecast origin $y_T$. For a collection of $N$ time series, let $i \in \{1, \ldots, N\}$ index the individual series. Across series directional accuracy is $\mathrm{DA}_{T,h} = \frac{1}{N}\sum_{i=1}^{N} d_{i,T,h}$, the mean of the series level indicators (Appendix C).

**Directional class labels.** At forecast origin $T$ and horizon $h$, define

$$z_{T,h} = \mathbb{I}\left[\mathrm{sign}(y_{T+h} - y_T) = 1\right], \tag{2}$$

where $z_{T,h} = 1$ denotes an increase from $y_T$ to $y_{T+h}$, and $z_{T,h} = 0$ denotes a non increase.

**Directional Prediction Framework.** The framework proceeds in four steps (Figure 1).

1. **Classifier predicted directional class label.** At forecast origin $T$ and horizon $h$, a time series classifier produces $\hat{z}_{T,h} \in \{0, 1\}$, where $\hat{z}_{T,h} = 1$ denotes a predicted increase relative to $y_T$, and $\hat{z}_{T,h} = 0$ denotes a predicted non increase.
2. **Forecast implied directional class label.** The point forecast implies

$$\hat{z}^{\mathrm{F}}_{T,h} = \mathbb{I}\left[\mathrm{sign}(\hat{y}_{T+h|T} - y_T) = 1\right]. \tag{3}$$

3. **Directional adjustment.** The adjustment factor is

$$\gamma_{T,h} = \begin{cases} 1, & \hat{z}_{T,h} = \hat{z}^{\mathrm{F}}_{T,h}, \\ \lambda_{\mathrm{Up}}, & \hat{z}_{T,h} \neq \hat{z}^{\mathrm{F}}_{T,h} \ \wedge \ \hat{z}_{T,h} = 1, \\ \lambda_{\mathrm{Down}}, & \hat{z}_{T,h} \neq \hat{z}^{\mathrm{F}}_{T,h} \ \wedge \ \hat{z}_{T,h} = 0, \end{cases} \tag{4}$$

   where $\lambda_{\mathrm{Up}} \geq 1$ and $0 < \lambda_{\mathrm{Down}} \leq 1$ are multiplicative adjustment parameters.
4. **Adjusted forecast.** The adjusted point forecast is

$$\tilde{y}_{T+h|T} = \gamma_{T,h}\hat{y}_{T+h|T}. \tag{5}$$

We use the adjustment in equation (5) as a post hoc exploratory proof of concept to assess whether classifier guidance on disagreement can improve point forecast accuracy. We use a multiplicative adjustment because M4 target values are positive.

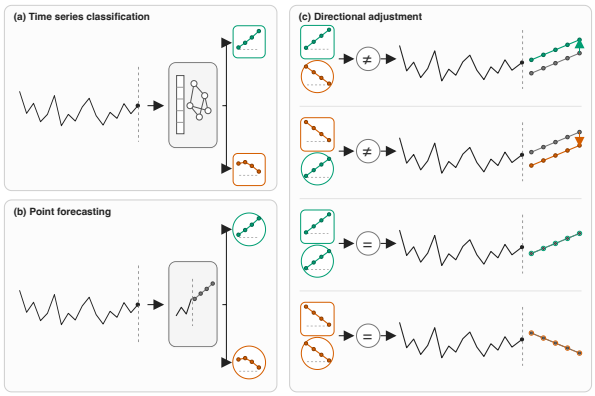



Figure 1: Directional Prediction Framework. Squares and circles denote classifier predicted and forecast implied labels, respectively; green and orange indicate increase and non increase. Agreement preserves the forecast; disagreement applies the classifier indicated adjustment.

## 4 Experiments

We evaluate the Directional Prediction Framework on the M4 dataset (Makridakis et al., 2018), comprising 100,000 time series across six frequencies and multiple forecast horizons. Following Section 3, we reformulate the forecasting tasks as directional prediction tasks. We evaluate point forecasting methods and time series classifiers, including foundation model based approaches, using directional accuracy. We compare the M4 winning SMYL (Smyl, 2020) and second ranked FFORMA (Montero-Manso et al., 2020) point forecasts with Chronos-2 (Ansari et al., 2025). The classifier comparison includes 1-NN DTW, 1-NN ED, Rotation Forest, XGBoost, ROCKET (Dempster et al., 2020), InceptionTime (Ismail Fawaz et al., 2020), and Mantis (Feofanov et al., 2025). The conventional classifiers span distance based, feature based, ensemble, kernel based, and deep learning approaches. Chronos-2 uses released weights without fine tuning but has documented M4 pretraining overlap. Mantis uses a frozen encoder with a 200 tree Random Forest. Implementation and dataset construction details are provided in Appendices C–D. Directional prediction is evaluated independently at each horizon. Tables 1 and 2 report directional accuracy at the terminal official horizon $h = H$, whereas sMAPE, MASE, and OWA are computed over the complete official forecast horizon. Forecast implied directional labels are calculated using equation (3). As a post hoc proof of concept, we select one global parameter pair per forecasting model and frequency by minimising full horizon OWA over 525 candidates on the official M4 evaluation set (Appendix B). Accordingly, adjusted results are interpreted as exploratory proof of concept estimates rather than held out performance estimates.

## 5 Results

Figure 2 descriptively shows how directional accuracy varies across horizons on the M4 Daily subset, with the highest observed values occurring at some longer horizons.

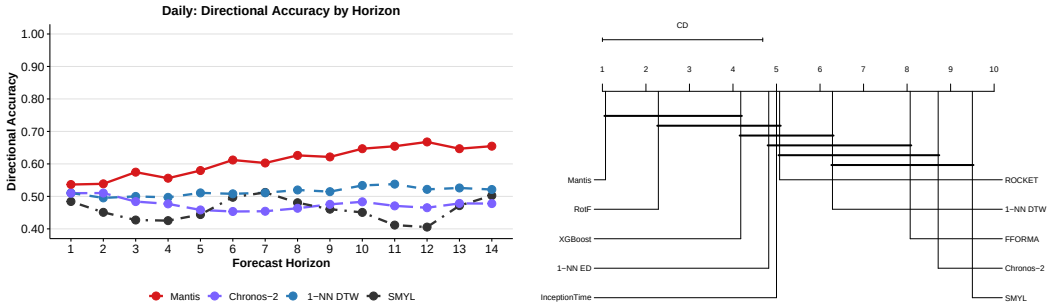



Figure 2: Directional prediction results for the M4 Daily subset. **Left:** Directional accuracy by forecast horizon. **Right:** Descriptive mean ranks across the 14 Daily forecast horizons, with lower ranks indicating better performance.

Mantis has the best average rank in Figure 2. A baseline that always predicts an increase achieves approximately $66\%$ accuracy pooled across Daily horizons (Appendix D, Figure 9). Across the remaining frequencies, directional profiles are heterogeneous, and no single method dominates (Appendix A). Additional details on the rank diagram and reproducibility are provided in Appendix D.

Table 1 compares the Daily forecasts with their Mantis guided adjustments, SMYL–Mantis and Chronos-2–Mantis. Relative to their unadjusted forecasts, the in sample Daily OWA reductions are $4.01\%$ for SMYL–Mantis and $5.52\%$ for Chronos-2–Mantis. The $7.64\%$ value in the table instead compares Chronos-2–Mantis with SMYL. The infeasible SMYL–Oracle achieves a substantially larger $47.34\%$ OWA improvement relative to SMYL, indicating considerable ex post headroom within the scaling construction rather than the performance of a feasible forecasting method.

Table 1: Daily forecasting performance. DA is reported at $h = 14$; sMAPE, MASE, and OWA are reported over all official horizons. Bold indicates the best feasible value in each column.

| **Method** | **sMAPE↓** | **MASE↓** | **DA↑** | **OWA↓** | **Improvement over SMYL (%)** | | |
|---|---|---|---|---|---|---|---|
| | | | | | **sMAPE↑** | **MASE↑** | **OWA↑** |
| Chronos-2 | 3.09 | 3.38 | 0.478 | 1.023 | 2.61 | 1.88 | 2.24 |
| FFORMA | 3.10 | 3.34 | 0.510 | 1.019 | 2.29 | 2.95 | 2.62 |
| SMYL | 3.17 | 3.45 | 0.502 | 1.046 | 0.00 | 0.00 | 0.00 |
| SMYL–Mantis | 3.05 | 3.30 | **0.670** | 1.004 | 3.69 | 4.33 | 4.01 |
| Chronos-2–Mantis | **2.95** | **3.16** | 0.665 | **0.966** | **6.86** | **8.41** | **7.64** |
| SMYL–Oracle | 1.72 | 1.76 | 0.859 | 0.551 | 45.64 | 49.02 | 47.34 |

## 6 Conclusion

We introduced a Directional Prediction Framework that enables point forecasting methods and time series classifiers to be evaluated using directional accuracy and uses learned directional signals to adjust point forecasts. On the M4 Daily subset, Mantis exhibits higher directional accuracy at some longer horizons, while directional accuracy profiles vary across frequencies. Mantis guided adjustments lower Daily OWA in sample after evaluation set parameter selection. Gains vary across settings and remain untested on unseen outcomes. These findings provide exploratory proof of concept evidence that learned directional representations can complement point forecasts. Time series foundation models therefore offer opportunities to study directional prediction as a complementary forecasting task. The evidence is limited to the M4 dataset and the methods considered. Future work should assess additional datasets and representations, investigate directional predictability, and learn adaptive adjustments with separate training, validation and evaluation data. All comparisons are descriptive and conditional on the reported experimental design (Appendix D).

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# A Forecasting Value of Directional Information

This appendix examines whether learned directional signals provide additional value for point forecasting across the six M4 frequencies. Figure 3 illustrates two directional adjustments when classifier predicted and forecast implied directions disagree. Table 2 summarises terminal horizon directional accuracy and full horizon point forecast accuracy across frequencies.

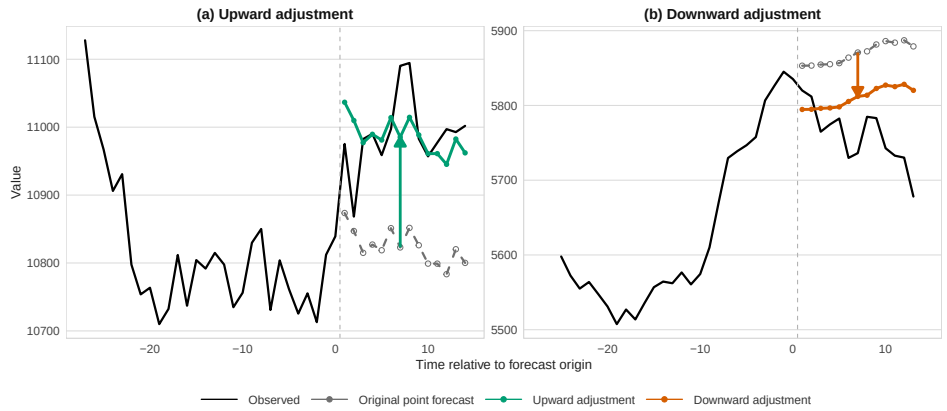



Figure 3: Directional adjustment examples from M4. Mantis adjusts the SMYL forecast upward for Daily series D1602 (left) and the Chronos-2 forecast downward for Weekly series W39 (right).

Table 2: Directional accuracy at the terminal official horizon and point forecast accuracy over the complete official forecast horizon, by M4 frequency.

**Panel A. Directional accuracy at $h = H$ (DA) ↑**

| Method | Hourly | Daily | Weekly | Monthly | Quarterly | Yearly | All |
|---|---|---|---|---|---|---|---|
| Chronos-2 | 0.785 | 0.478 | 0.596 | 0.549 | 0.532 | 0.542 | 0.542 |
| FFORMA | 0.688 | 0.510 | 0.649 | 0.694 | 0.706 | 0.765 | 0.705 |
| SMYL | 0.761 | 0.502 | 0.635 | **0.722** | 0.717 | 0.784 | 0.726 |
| SMYL–Mantis | 0.766 | **0.670** | 0.652 | **0.722** | **0.721** | **0.786** | **0.734** |
| Chronos-2–Mantis | **0.809** | 0.665 | **0.708** | 0.665 | 0.693 | 0.725 | 0.686 |
| Mantis | 0.643 | 0.655 | 0.596 | 0.522 | 0.700 | 0.778 | 0.630 |
| SMYL–Oracle | 0.829 | 0.859 | 0.841 | 0.859 | 0.861 | 0.909 | 0.871 |

**Panel B. Overall Weighted Average (OWA) ↓**

| Method | Hourly | Daily | Weekly | Monthly | Quarterly | Yearly | All |
|---|---|---|---|---|---|---|---|
| Chronos-2 | 0.393 | 1.023 | 0.712 | 0.888 | 0.911 | 0.891 | 0.895 |
| FFORMA | 0.484 | 1.019 | 0.796 | 0.858 | 0.847 | 0.799 | 0.838 |
| SMYL | 0.440 | 1.046 | 0.851 | **0.836** | 0.847 | 0.778 | 0.821 |
| SMYL–Mantis | 0.438 | 1.004 | 0.839 | **0.836** | **0.847** | **0.778** | **0.819** |
| Chronos-2–Mantis | **0.390** | **0.966** | **0.695** | 0.888 | 0.898 | 0.864 | 0.880 |
| SMYL–Oracle | 0.356 | 0.551 | 0.540 | 0.557 | 0.502 | 0.383 | 0.479 |

**Panel C. OWA improvement relative to SMYL (%) ↑**

| Method | Hourly | Daily | Weekly | Monthly | Quarterly | Yearly | All |
|---|---|---|---|---|---|---|---|
| Chronos-2 | 10.78 | 2.24 | 16.29 | -6.29 | -7.57 | -14.51 | -9.01 |
| FFORMA | -9.92 | 2.62 | 6.49 | -2.65 | 0.03 | -2.69 | -2.05 |
| SMYL | 0.00 | 0.00 | 0.00 | **0.00** | 0.00 | 0.00 | 0.00 |
| SMYL–Mantis | 0.57 | 4.01 | 1.42 | **0.00** | **0.07** | **0.02** | **0.25** |
| Chronos-2–Mantis | **11.31** | **7.64** | **18.27** | -6.23 | -6.00 | -11.02 | -7.19 |
| SMYL–Oracle | 19.03 | 47.34 | 36.52 | 33.40 | 40.69 | 50.76 | 41.61 |

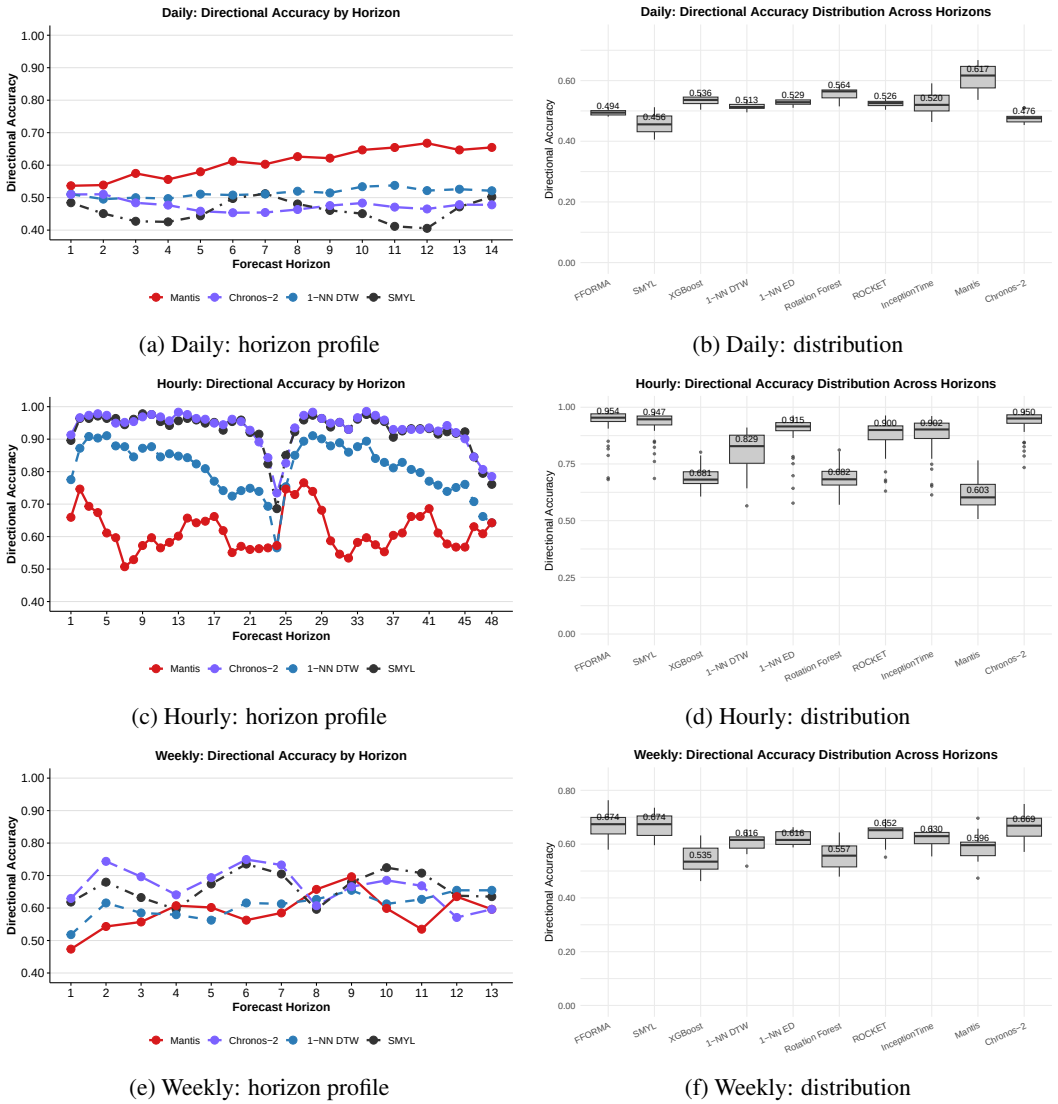
*Notes.*

(i) Bold indicates the best feasible value in each column; Panel C percentages use unrounded OWA values.
(ii) Terminal $H$ for Hourly through Yearly is 48, 14, 13, 18, 8, and 6, respectively.
(iii) All DA is weighted by series count; All OWA is recomputed from weighted sMAPE and MASE relative to Naive2.
(iv) Adjusted results use parameters selected on the evaluation set and are exploratory; SMYL Oracle uses realised outcomes and is infeasible.

The Daily subset offers the clearest demonstration. Mantis based adjustment increases Chronos-2 terminal horizon directional accuracy from $0.478$ to $0.665$ and reduces OWA from $1.023$ to $0.966$. The same adjustment increases SMYL terminal horizon directional accuracy from $0.502$ to $0.670$ and reduces OWA from $1.046$ to $1.004$. These adjusted OWA results use the post hoc parameter grid analysis described in Appendix B and are therefore exploratory proof of concept estimates rather than held out performance estimates.

The value of directional guidance varies across frequencies and base forecasters. Within model OWA reductions are largest on Daily and negligible on Monthly. Mantis has lower aggregate terminal horizon directional accuracy than SMYL, yet SMYL–Mantis yields a small aggregate OWA reduction.

Figures 2, 4, and 5 show the corresponding horizon structure. On the Daily subset, Mantis directional accuracy increases from $0.537$ at the first horizon to $0.668$ at horizon 12; the remaining frequencies exhibit more setting specific profiles. These profiles need not follow the corresponding patterns of point forecast accuracy.




(a) Daily: horizon profile

(b) Daily: distribution

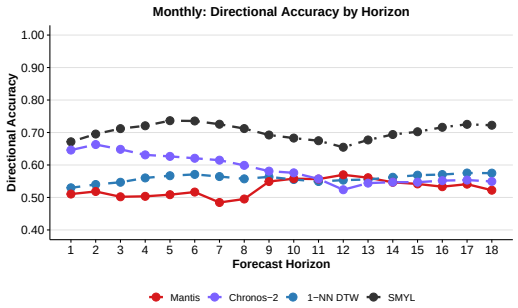
(c) Hourly: horizon profile

(d) Hourly: distribution

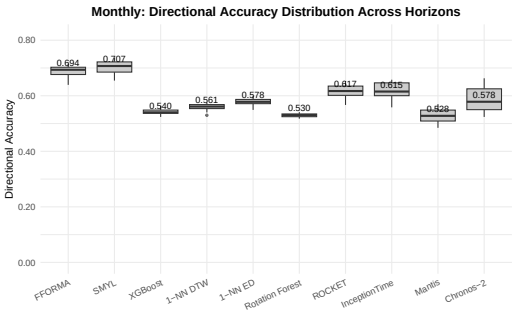
(e) Weekly: horizon profile

(f) Weekly: distribution

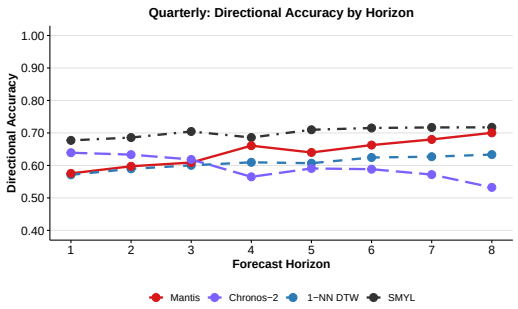
Figure 4: Directional accuracy across forecast horizons for the M4 Daily, Hourly, and Weekly subsets. **Left:** horizon profiles for selected methods. **Right:** box plots for all methods; labels report medians.



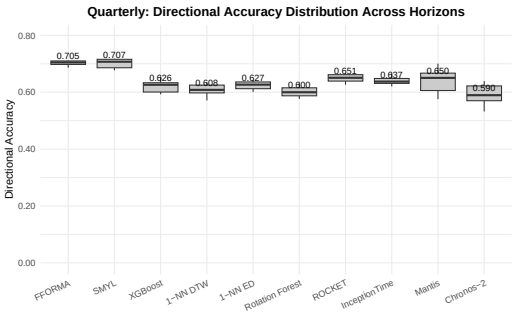

(a) Monthly: horizon profile



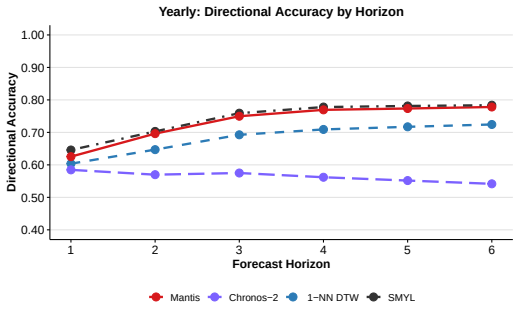

(b) Monthly: distribution



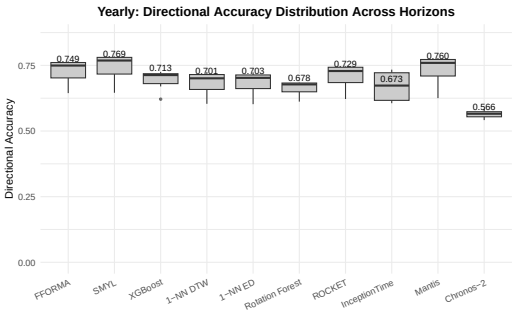

(c) Quarterly: horizon profile




(d) Quarterly: distribution




(e) Yearly: horizon profile




(f) Yearly: distribution

Figure 5: Directional accuracy across forecast horizons for the M4 Monthly, Quarterly, and Yearly subsets. **Left:** horizon profiles for selected methods. **Right:** box plots for all methods; labels report medians.

The conventional classifiers learn the same horizon specific targets as Mantis, but do not match its average Daily directional rank. This motivates investigating pretrained classification representations, beyond the decision to learn direction directly.

The contrasting Daily profiles of Mantis and Chronos-2 also show that foundation model status alone does not imply similar directional behaviour. These comparisons do not isolate pretraining as the cause of the observed differences.

Different prediction objectives offer one possible explanation: minimising numerical forecast error need not maximise directional accuracy, while predicting direction alone provides no estimate of the future magnitude.

SMYL–Oracle is the infeasible ex post scaling diagnostic. For each series, it selects the multiplier $m \in \{0.500, 0.501, \ldots, 1.500\}$ that minimises SMYL sMAPE over that series' realised official forecast horizon. On the Daily subset, it reduces OWA by $47.34\%$ relative to SMYL. The gap between this infeasible result and the feasible SMYL–Mantis result indicates headroom within the scaling construction when realised outcomes are available, but it should not be attributed solely to unexploited directional information.

Together, the results provide exploratory evidence that learned directional representations can complement point forecasts in suitable settings. The Directional Prediction Framework makes this contribution measurable across point forecasting methods, time series classifiers, and future foundation model representations. Transfer beyond M4 and adjustment using other classifiers remain untested.

## B Sensitivity Analysis of Directional Adjustment

This appendix assesses whether directional adjustment gains are robust to variation in the adjustment parameters. For each forecasting model and frequency, we evaluated 525 parameter pairs, with $\lambda_{\mathrm{Up}} \in [1.00, 1.12]$ and $\lambda_{\mathrm{Down}} \in [0.90, 1.00]$, both in increments of 0.005. When the Mantis and forecast implied directional labels disagreed at the terminal horizon, we applied the multiplier specified by the Mantis label to the point forecast. Agreement left the path unchanged. We selected the pair with the lowest full horizon OWA on the official M4 evaluation set and used it for all series within that frequency. Because we used the same outcomes to select parameters and measure improvement, the reported gains are in sample. The gains provide proof of concept evidence that Mantis derived signals can complement point forecasts through a global adjustment. The optimal adjustment rule and performance on unseen outcomes remain open questions. Figure 6 shows the Daily response surfaces for both adjusted forecasts.

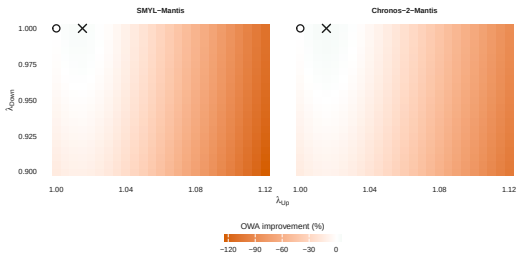



Figure 6: Directional adjustment sensitivity for the M4 Daily subset. Colour shows OWA improvement relative to the corresponding unadjusted forecast. Circles mark no adjustment, $(\lambda_{\mathrm{Up}}, \lambda_{\mathrm{Down}}) = (1, 1)$; crosses mark the minimum OWA setting in each panel.

The Daily response surfaces are smooth in the vicinity of the minimum OWA settings. Performance varies more strongly with $\lambda_{\mathrm{Up}}$ than with $\lambda_{\mathrm{Down}}$, but improvements occur over local regions rather than at isolated parameter pairs. Table 3 summarises the minimum OWA settings and robustness measures across frequencies.

Table 3: Directional adjustment settings and robustness across M4 frequencies.

**Panel A. Minimum OWA parameter settings**

| Frequency | Adjusted method | $\lambda^*_{\mathrm{Up}}$ | $\lambda^*_{\mathrm{Down}}$ | Unadjusted OWA ↓ | Minimum OWA ↓ | OWA improvement (%) ↑ |
|---|---|---|---|---|---|---|
| Hourly | Chronos-2–Mantis | 1.005 | 1.000 | 0.393 | **0.390** | **0.60** |
| Hourly | SMYL–Mantis | 1.005 | 1.000 | 0.440 | 0.438 | 0.57 |
| Daily | Chronos-2–Mantis | 1.015 | 1.000 | 1.023 | **0.966** | **5.52** |
| Daily | SMYL–Mantis | 1.015 | 1.000 | 1.046 | 1.004 | 4.01 |
| Weekly | Chronos-2–Mantis | 1.015 | 0.990 | 0.712 | **0.695** | **2.38** |
| Weekly | SMYL–Mantis | 1.015 | 0.995 | 0.851 | 0.839 | 1.42 |
| Monthly | Chronos-2–Mantis | 1.005 | 1.000 | 0.888 | 0.888 | **0.06** |
| Monthly | SMYL–Mantis | 1.000 | 1.000 | 0.836 | **0.836** | 0.00 |
| Quarterly | Chronos-2–Mantis | 1.025 | 1.000 | 0.911 | 0.898 | **1.45** |
| Quarterly | SMYL–Mantis | 1.005 | 1.000 | 0.847 | **0.847** | 0.07 |
| Yearly | Chronos-2–Mantis | 1.065 | 0.995 | 0.891 | 0.864 | **3.05** |
| Yearly | SMYL–Mantis | 1.010 | 1.000 | 0.778 | **0.778** | 0.02 |

**Panel B. Robustness over the parameter grid**

| Frequency | Adjusted method | $\lambda_{\mathrm{Up}}$ range | $\lambda_{\mathrm{Down}}$ range | Improving grid points (%) ↑ |
|---|---|---|---|---|
| Hourly | Chronos-2–Mantis | 1.000–1.005 | 1.000–1.000 | 0.19 |
| Hourly | SMYL–Mantis | 1.000–1.010 | 0.995–1.000 | **0.38** |
| Daily | Chronos-2–Mantis | 1.010–1.020 | 0.985–1.000 | **9.33** |
| Daily | SMYL–Mantis | 1.010–1.015 | 0.985–1.000 | 6.67 |
| Weekly | Chronos-2–Mantis | 1.005–1.025 | 0.980–1.000 | **7.81** |
| Weekly | SMYL–Mantis | 1.005–1.035 | 0.985–1.000 | 6.48 |
| Monthly | Chronos-2–Mantis | 1.000–1.015 | 0.975–1.000 | **0.38** |
| Monthly | SMYL–Mantis | 1.000–1.010 | 0.980–1.000 | 0.00 |
| Quarterly | Chronos-2–Mantis | 1.005–1.045 | 0.900–1.000 | **35.43** |
| Quarterly | SMYL–Mantis | 1.000–1.030 | 0.910–1.000 | 0.95 |
| Yearly | Chronos-2–Mantis | 1.030–1.110 | 0.905–1.000 | **94.48** |
| Yearly | SMYL–Mantis | 1.000–1.085 | 0.915–1.000 | 1.14 |

*Notes.*

(i) Bold indicates the better of the two methods within each frequency and outcome column.
(ii) OWA improvement is $100(1 - \mathrm{OWA}_{\mathrm{adjusted}}/\mathrm{OWA}_{\mathrm{unadjusted}})$ from unrounded values; improving grid points is the percentage of 525 pairs with OWA below the corresponding unadjusted value.
(iii) Parameter ranges are marginal extrema among grid points no more than 1% above minimum OWA; points inside the implied rectangle need not meet this criterion.
(iv) Results use parameters selected on the official M4 evaluation set and are exploratory, not held out estimates.

OWA improvement is reported relative to each method's corresponding unadjusted forecast. For Daily, the reductions are $4.01\%$ for SMYL–Mantis relative to SMYL and $5.52\%$ for Chronos-2–Mantis relative to Chronos-2. Daily shows the largest within model improvements, whereas Monthly adjustments provide negligible gains. Quarterly and Yearly Chronos-2–Mantis adjustments have the broadest improving regions but more modest best improvements. Thus, attainable gain and the extent of the parameter region over which improvement occurs are distinct properties. Figure 7 shows the corresponding response surfaces across frequencies and base forecasts.

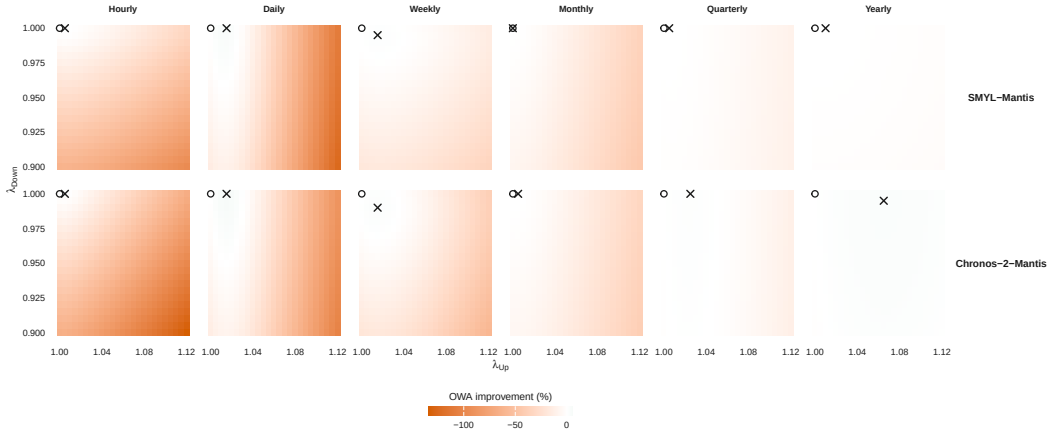



Figure 7: Directional adjustment sensitivity across M4 frequencies and base forecasts. Colour shows OWA improvement relative to the corresponding unadjusted forecast. Circles mark no adjustment, $(\lambda_{\mathrm{Up}}, \lambda_{\mathrm{Down}}) = (1, 1)$; crosses mark the minimum OWA setting in each panel.

The shape and extent of the improving regions differ across frequencies and base forecasts. Surfaces remain smooth throughout the evaluated grid. At the selected Daily setting $(\lambda_{\mathrm{Up}}, \lambda_{\mathrm{Down}}) = (1.015, 1.000)$, Mantis disagrees with SMYL and Chronos-2 on 46.94% and 38.06% of the series, respectively. Among these disagreements, the adjusted terminal forecast adopts the Mantis label in $86.90\%$ and $76.88\%$ of cases, respectively, while OWA decreases by $8.48\%$ and $13.31\%$ within the corresponding disagreement subsets. Because $\lambda_{\mathrm{Down}} = 1$, forecasts associated with downward disagreements remain unchanged.

Future work should compare Mantis guided adjustment with the unadjusted forecast, an upward correction and adjustment guided by a basic classifier under a common held out protocol. These controls would help distinguish the value of Mantis's guidance from the correction itself.

## C Directional Prediction Dataset Construction

Section 3 defines the directional labels and evaluation criterion. Forecasting datasets such as M4 are not naturally structured as directional prediction tasks. This appendix operationalises the framework by converting unequal length M4 series into fixed length, labelled instances for horizon specific binary classification. The resulting datasets follow the labelled series conventions used in the UCR Archive and TSC Bake Off studies (Dau et al., 2019; Bagnall et al., 2017). Point forecasting methods retain their native forecast generation procedures.

**Directional accuracy across series.** For a method applied to a dataset containing $N$ time series, let $i \in \{1, \ldots, N\}$ denote the index of an individual series. Directional accuracy at forecast origin $T$ and horizon $h$ is defined as

$$\mathrm{DA}_{T,h} = \frac{1}{N} \sum_{i=1}^{N} d_{i,T,h}, \tag{6}$$

where $d_{i,T,h}$ denotes the directional accuracy indicator in equation (1) for time series $i$.

Figure 8 summarises the rolling window construction and resulting time series classification tasks.

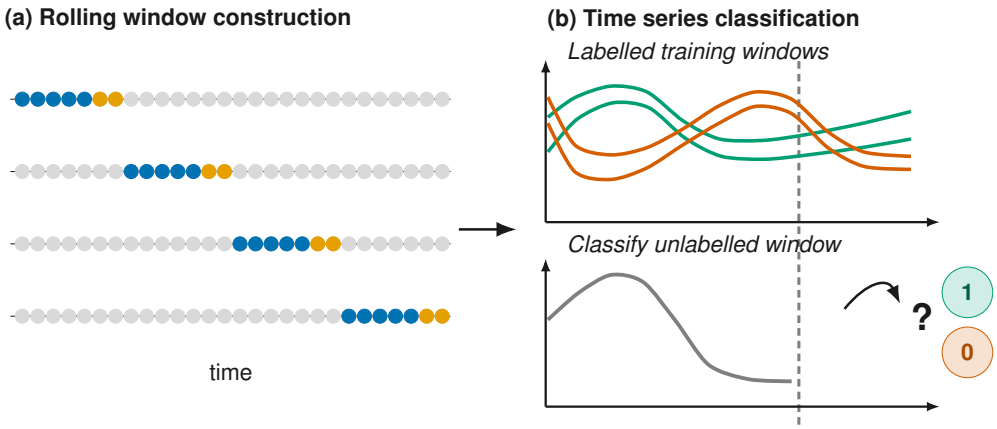



Figure 8: Construction of horizon specific directional prediction datasets. **(a)** Blue points form fixed length classifier inputs, orange points are future observations used to assign labels, and grey points lie outside each instance. **(b)** At horizon $h$, labelled windows train a binary classifier that predicts increase ($z_{T,h} = 1$) or non increase ($z_{T,h} = 0$) for a new window.

**Construction of directional prediction tasks.** Consider one rolling window of length $w$, ending at forecast origin $T$:

$$\mathbf{y}_{T,w} = (y_{T-w+1}, \ldots, y_T).$$

The procedure is repeated across rolling windows and time series.

1. **Construct rolling windows.** Following the rolling origin principle described by Hyndman and Athanasopoulos (2021), fixed length windows are generated using window length $w$, maximum forecast horizon $H$, and step size $w + H$. For the reported results, $w = (256, 64, 64, 64, 32, 16)$ for Hourly, Daily, Weekly, Monthly, Quarterly, and Yearly series, respectively.
2. **Standardise the rolling windows.** Let $\mu$ and $\sigma$ be the mean and standard deviation of $\mathbf{y}_{T,w}$. If $\sigma > 0$, each observation is standardised as

   $$y_t^{\mathrm{std}} = \frac{y_t - \mu}{\sigma}, \qquad t = T - w + 1, \ldots, T.$$

   Applying this transformation elementwise gives the standardised window $\mathbf{y}_{T,w}^{\mathrm{std}}$, which is the classifier input. If $\sigma = 0$, $\mathbf{y}_{T,w}^{\mathrm{std}}$ is set to the zero vector.
3. **Assign directional class labels.** For each forecast horizon $h = 1, \ldots, H$, the window is assigned the directional label $z_{T,h}$ defined in equation (2). The future observation $y_{T+h}$ is used only to assign the label and is not included in the classifier input.

4. **Fit the classifiers.** Series yielding at least one complete length $w$ input window followed by $H$ outcomes were randomly assigned to training and internal hold out partitions using an 80:20 series level split. Ineligible series were omitted from classifier training without padding or imputation, and all windows from a series remained in the same partition. A separate classifier was fitted for each horizon using only windows from the training series.

This construction converts unequal length forecasting series into fixed length, scale standardised, horizon specific directional prediction datasets while preserving the information available at each forecast origin. It thereby enables time series classifiers and point forecasting methods to be evaluated on a common directional task.

## D Experimental Design and Reproducibility

**Dataset characteristics.** Experiments use the M4 dataset (Makridakis et al., 2018). Table 4 reports the number of series by frequency and domain. Figure 9 reports directional class proportions pooled across all series and official forecast horizons within each frequency, using the labels defined in equation (2).

Table 4: Number of M4 series by frequency and domain.

| Frequency | Micro | Industry | Macro | Finance | Demographic | Other | Total |
|---|---|---|---|---|---|---|---|
| Yearly | 6,538 | 3,716 | 3,903 | 6,519 | 1,088 | 1,236 | 23,000 |
| Quarterly | 6,020 | 4,637 | 5,315 | 5,305 | 1,858 | 865 | 24,000 |
| Monthly | 10,975 | 10,017 | 10,016 | 10,987 | 5,728 | 277 | 48,000 |
| Weekly | 112 | 6 | 41 | 164 | 24 | 12 | 359 |
| Daily | 1,476 | 422 | 127 | 1,559 | 10 | 633 | 4,227 |
| Hourly | 0 | 0 | 0 | 0 | 0 | 414 | 414 |
| Total | 25,121 | 18,798 | 19,402 | 24,534 | 8,708 | 3,437 | 100,000 |

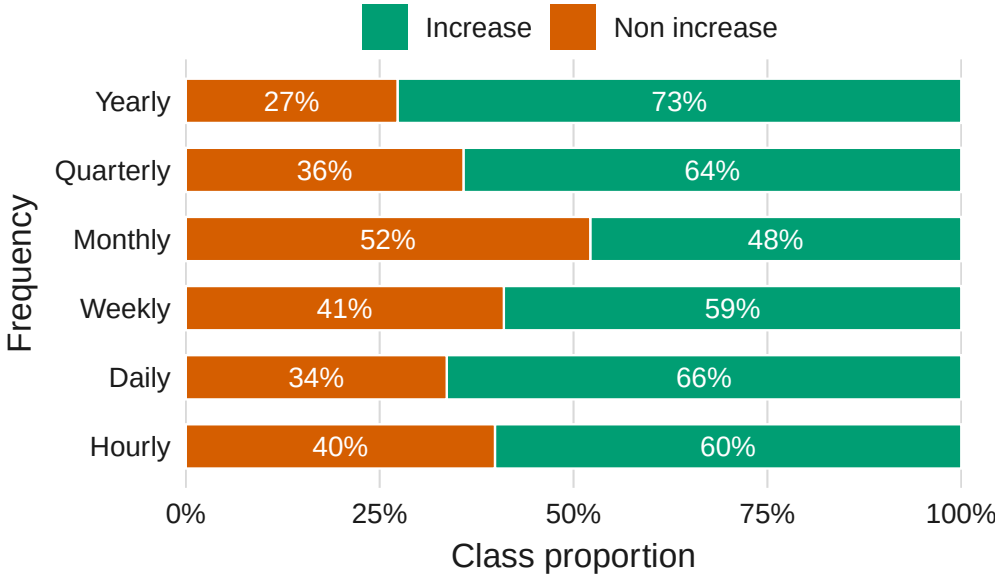



Figure 9: Directional class proportions pooled across all series and official forecast horizons within each M4 frequency. Increase denotes $y_{T+h} > y_T$; non increase denotes $y_{T+h} \leq y_T$, for $h = 1, \ldots, H$, where $y_T$ is the final observed value.

**Class prior reference.** Always predicting the more frequent class within a frequency achieves the larger proportion in Figure 9. This is a descriptive reference based on pooled evaluation outcomes.

**Training and evaluation protocol.** Dataset construction, including the series level split, training window eligibility, and classifier fitting, is described in Appendix C. At each horizon, all M4 series were evaluated at their official final forecast origins, including training partition and held out series. Their official future outcomes were not used as classifier training labels. Directional accuracy follows equations (1) and (6). These results are descriptive; separate estimates for held out series and series level uncertainty intervals are not reported.

**Model configuration.** SMYL and FFORMA forecasts were obtained from `M4comp2018` (Montero-Manso et al., 2018). Chronos-2 used the publicly available `amazon/chronos-2` checkpoint without task specific fine tuning. Its pretraining data include the M4 Daily, Hourly, Monthly, and Weekly subsets (Ansari et al., 2025); results for these frequencies therefore involve pretraining overlap.

Mantis followed the authors' frozen backbone feature extraction protocol (Feofanov et al., 2025): representations were extracted without fine tuning and classified using a 200 tree Random Forest with unrestricted maximum depth. The pretraining corpus reported for Mantis does not include M4; therefore, unlike Chronos-2, Mantis has no documented direct M4 pretraining overlap. ROCKET, InceptionTime, 1-NN DTW, 1-NN ED, and Rotation Forest used their standard `sktime` implementations (Löning et al., 2019). For each frequency, window setting, and horizon, XGBoost used a binary logistic objective; its eight hyperparameters were selected by five fold log loss Bayesian optimisation with 8 initial and 20 sequential evaluations, at most 1,500 boosting rounds, and 15 round early stopping, following the FFORMA tuning approach (Montero-Manso et al., 2020). Directional adjustment follows equations (4) and (5), using the post hoc parameter values reported in Appendix B.

**Model comparison.** The right panel of Figure 2 was generated using `scmamp::plotCD` with $\alpha = 0.05$ (Calvo and Santafé, 2016) and reports mean ranks across the 14 Daily forecast horizons. Because all horizons are evaluated on the same Daily series, the critical difference bar and grouping lines are not interpreted as evidence of statistical significance. Only the method positions on the rank axis are interpreted descriptively.

**Compute resources.** The experiments were executed primarily on a local workstation running Ubuntu 24.04 LTS, equipped with an AMD Ryzen 9 3950X CPU (16 cores and 32 threads), 128 GB of RAM, and one NVIDIA GeForce RTX 5090 GPU with 32 GB of VRAM. Exact wall clock times and the additional compute consumed by preliminary experiments were not systematically recorded; runtimes may also vary with software configuration and available parallelism.

**Reproducibility.** The public repository contains the complete pipeline in R and Python for dataset construction, fixed series level splits, model fitting and evaluation, Chronos-2 inference, sensitivity analysis, and regeneration of the reported tables and figures. The scripts record the seeds, window settings, model hyperparameters, checkpoint identifiers, and adjustment grid.

The README documents installation and the exact execution order. R package lockfiles and Conda environment definitions specify the software environments, while generated artifacts use fixed project paths.

**Asset licences.** The accompanying code developed for this study is released under the MIT License. The M4 data are accessed through `M4comp2018` version 0.2.0 (GPL-3); the `amazon/chronos-2` and `paris-noah/Mantis-8M` checkpoints are released under Apache-2.0; `sktime` is released under BSD-3-Clause; `XGBoost` is released under Apache-2.0; and `scmamp` is released under GPL-2. Exact software versions and checkpoint identifiers are recorded in the accompanying repository. These assets are credited to their original authors, used under their respective terms, and downloaded from their official sources rather than redistributed.

**Code availability.** The research code and environment definitions are publicly available at `https://github.com/rafmontano/m4_tsc_fmts_2026`.

**Broader Impacts.** Directional forecasting may improve decision support in domains where anticipating the direction of change is more important than predicting its precise magnitude. However, incorrect directional predictions or deployment beyond the evaluated M4 setting could lead to harmful financial or operational decisions. Domain specific validation, uncertainty assessment, monitoring, and human oversight are therefore necessary, particularly in high stakes applications.